

Modules and Group Representations

(MATH 122)

Lecture Notes

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Disclaimer:

These are the notes I took for Math 122 in Spring 2026 at Stanford. The course was taught by Prof. Daniel Bump, but any errors and misconception are almost definitely mine. I also did add a lot of notes that I took from various snippets of readings I did through the course, especially from Dummit and Foote's *Abstract Algebra*, Lang's *Algebra*, and random notes and videos on the web, and rephrased a lot of arguments in my own understanding of them.

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Prerequisites

Theorem 1 (Zorn's Lemma). *Let S be a partially ordered set. If every totally ordered subset of S has an upper bound in S , then S contains a maximal element.*

This paper goes deeply into Zorn's lemma and some of its applications.

Basic Facts About A_n

1. When $n \geq 3$, A_n is generated by the set of all 3 cycles
2. When $n \geq 3$, the commutator subgroup of S_n , denoted S'_n is A_n : This is a consequence of the above and the above fact and the fact that every commutator element must be even. We could also use the fact that $[S_n : A_n] = 2 \implies A_n$ is abelian and thus $S'_n \leq A_n$
3. When $n \geq 5$, A_n is simple, so S_n is not solvable
4. When $n \leq 3$, A'_n is trivial, when $n = 4$, it is V_4 and when $n \geq 5$, $A'_n = A_n$

Basic Facts About D_{2n}

1. The commutator subgroup of D_{2n} is generated by r^2 , so

$$D'_{2n} = \langle r \rangle$$

$$2. \quad D_{2n}/D'_{2n} = \begin{cases} Z_2 & \text{if } n \bmod 2 = 1 \\ Z_2 \times Z_2 & \text{if } n \bmod 2 = 0 \end{cases}$$

This is pretty easy to see from the previous point but is pretty useful to keep in mind!

3. For $n \geq 3$, D_{2n} is non-abelian
4. **Conjugacy:** If n is even, the reflections form one conjugacy class. If n is odd, the reflections split into two conjugacy classes (those fixing vertices and those fixing sides). In all cases, pairs of inverse rotations are conjugate.

Trace and Direct Sum Decompositions

Let W be a finite-dimensional vector space with a direct sum decomposition

$$W = W_1 \oplus W_2 \oplus \cdots \oplus W_m.$$

If $T : W \rightarrow W$ is a linear map, then relative to this decomposition T can be written as a block matrix

$$T = \begin{pmatrix} T_{11} & T_{12} & \cdots & T_{1m} \\ T_{21} & T_{22} & \cdots & T_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ T_{m1} & T_{m2} & \cdots & T_{mm} \end{pmatrix},$$

where

$$T_{ij} : W_j \rightarrow W_i.$$

The trace of T only depends on the diagonal blocks:

$$\text{tr}(T) = \sum_{i=1}^m \text{tr}(T_{ii}).$$

In particular, if T sends a summand W_i entirely into a different summand W_j with $j \neq i$, then that part of T contributes nothing to the trace. Equivalently, trace only sees the components of a linear map that return each summand to itself.

Modules

Definition 2 (Module). A (left) module over a ring R is an abelian group $(M, +)$ along with a multiplication $R \times M \rightarrow M$ which satisfies associative and distributive laws, i.e. for all $r, r_1, r_2 \in R$ and $m, m_1, m_2 \in M$, we have

- $r_1(r_2m) = (r_1r_2)m$

- $r(m_1 + m_2) = rm_1 + rm_2$

If M satisfies the above, we often just say that M is an R -module, or plainly a module if the base ring R is obvious from context.

- Examples 3.**
1. We can view modules as a generalization of vector spaces (in which case the base ring is a field, so we can divide by ring elements).
 2. We can also view them as a generalization of abelian groups: every abelian group is a module over $\mathbb{Z}$ through the multiplication $zm = m + \cdots + m$ (m times).
 3. The ring R can be viewed as a module over itself. The “left regular module” of R is often denoted ${}_R R$ to make the distinction.
 4. Column vectors in R^n are modules over the ring $\text{Mat}_n(R)$.

We can then define submodules in a similar way to how subspaces of vector spaces are defined.

Definition 4 (Submodule). Let M be an R -module. A submodule S of M is a subset $S \subseteq M$ that contains 0 and for which $r_1 s_1 + r_2 s_2 \in S$ for all $s_1, s_2 \in S$ and $r_1, r_2 \in R$. It is closed under “vector addition” and “multiplication by scalars”.

- Examples 5.**
1. The submodules of the left regular module of a ring R are exactly its left ideals.
 2. If R is ID, any R -module M has a submodule

$$\text{Tors}(M) = \{m \in M \mid \exists r \neq 0 \in R : rm = 0\}$$

called the torsion submodule.

- If $M = \text{Tors}(M)$, we say that M is torsion.
- If $0 = \text{Tors}(M)$, we say that M is torsion-free.

We can think of the torsion submodule as these vectors with “finite order”. Geometrically, it’s like if whenever we multiplied these vectors “enough times”, they eventually wrapped around and went back to 0.

Definition 6 (Simple Module). An R -module is called simple if it is not the zero module and its only submodules are $\{0\}$ and itself.

Theorem 7 (Schur’s Lemma). A homomorphism between simple R -modules is either zero or an isomorphism. This means that $\text{End}_R M$ is a division ring as every nonzero element has an inverse.

This post gives a version of Schur’s Lemma (with more assumptions) that is just great for understanding the power of it.

Proof. Let ϕ be a homomorphism of simple modules. The theorem is not complicated to see when we realize that $\ker \phi$ and $\text{Im } \phi$ must be submodules of their given modules. $\square$

Theorem 8 (Semisimplicity). *Let M be an R -module. Then the following properties are equivalent:*

1. M is the sum of a family of simple submodules.
2. M is the direct sum of a family of simple submodules.
3. For every submodule N , there exists a complementary submodule N' such that

$$M = N \oplus N'.$$

If M satisfies these equivalent conditions, it is called semisimple.

Proof taken from this.

Proof.

1 $\implies$ 2 Say $M = \sum_{i \in I} M_i$. Pick the maximal subfamily $\{M_j\}_{j \in J \subseteq I}$ for which the sum is direct, which exists from Zorn's lemma. Then consider

$$M' = \bigoplus_{j \in J} M_j$$

Notice that $M' \cap M_i$ is a submodule of M_i , so either equals 0 or M_i by Schur's lemma. However if this intersection is 0, then M' could be extended to include M_i which would contradict maximality, so M' contains every M_i .

2 $\implies$ 3 Apply the same argument as above but with Zorn's lemma used on the set of direct sums of submodules for which $M_i \cap N = \{0\}$. Then we get $M = N \oplus N'$ for some N' which is a direct sum of simple submodules.

3 $\implies$ 1 Let N be the sum of all simple submodules of M . By assumption, $M = N \oplus N'$ for some submodule N' . So it is enough to show that any nonzero submodule of M contains a simple submodule.

Let $N' \neq 0$ be a submodule of M , and choose $0 \neq v \in N'$. By Zorn's lemma, there is a maximal submodule $L \subsetneq Rv$. Since M is completely reducible, we have $M = L \oplus S$ for some submodule S . Intersecting with Rv gives

$$Rv = L \oplus (S \cap Rv).$$

Set $S' = S \cap Rv$. Then $S' \neq 0$, since otherwise $Rv = L$. Also S' is simple, because any submodule of S' gives a submodule of Rv containing L , so by maximality of L it must be either 0 or all of S' . Thus every nonzero submodule of M contains a simple submodule, so $N' = 0$. Therefore $M = N$, so M is the sum of its simple submodules. $\square$

Representations

Definition 9. A free module of rank n is constructed over the ring R from any abstract set $T = \{t_1, \dots, t_n\}$ by simply taking all formal linear combinations of the elements of T with coefficients in R and defining addition and multiplication by elements of R in the obvious manner. It is often noted $R[T]$ or $R\langle T \rangle$.

Definition 10 (Group Rings). If R is a ring and G a group, the group ring $R[G]$ is the free R -module on basis $\{g : g \in G\}$ with multiplication extended R -bilinearly from the group law

$$\left(\sum_g a_g g\right)\left(\sum_h b_h h\right) = \sum_{g,h} a_g b_h (gh)$$

where only finitely many coefficients are allowed to be nonzero.

Theorem 11 (Equivalence of Representations and Modules over Group Rings). *There are two main ways to view the concept of a representation of a group:*

- A representation of a group G over a field F is a vector space V over F together with an action of each group element $g \in G$ as a linear map $V \rightarrow V$ in a way that respects the group law. In short, it is a homomorphism $\rho : G \rightarrow GL(V)$.
- Define $F[G]$, the group ring, to be the structure containing all formal linear combinations of group elements in G with coefficients in F . A representation V of a group G over a field F is a $F[G]$ -module.

With this mind, simple modules correspond to what we call irreducible representations.

Proof. The correspondence is not too hard to see:

- When multiplying elements of $F[G]$ by elements of V , we can apply ρ and use the fact that it is a homomorphism and composition to show that multiplication is well defined.
- In the other direction, every $F[G]$ -module also defines a representation of G .

□

Definition 12 (Intertwiner). If ρ_V and ρ_W are representations of a group G , then a map $T : V \rightarrow W$ is an intertwiner (also called equivariant map) if it is a homomorphism of these representations i.e.

$$T\rho_V = \rho_W T.$$

This notion can also be used to talk about linear maps that respects the group structure for a specific group action:

$$T\rho(g) = \rho(g)T \iff \rho(g)^{-1}T\rho(g)T \quad \forall g \in G$$

For more information, see this.

Theorem 13 (Maschke's Theorem). *If F is a field and $\text{char } F \nmid |G|$ for a group G then $F[G]$ is semisimple.*

More specifically, let $\pi : G \rightarrow GL(V)$ be a representation of a group G over a F -vector space with $\text{char } F \nmid |G|$, then if W is an invariant subspace of V , we can find another invariant subspace W' such that $V = W \oplus W'$.

Proof. The proof starts by finding some idempotent $e : V \rightarrow W$ which must exist since $V = W \oplus X$ for some subspace X by usual linear algebra. Then, we consider the averaging map

$$\chi : V \rightarrow V \quad \chi(v) = \frac{1}{|G|} \sum_{g \in G} (g^{-1}eg)(v)$$

There are a few nice properties of χ that are easily verified:

- If $w \in W : \chi(w) = w$ so $\chi|_W = Id_W$
- If $v \in W, \chi(v) \in W$
- $\chi \in \text{End}(\rho)$ is an intertwiner for the representation.

From there, we can write $V = \text{Im}(\chi) + \ker(\chi)$ where $\ker(\chi)$ is invariant subspace of V as needed (since $\chi(gk) = g\chi(k) = 0$ if $k \in \ker(\chi)$). $\square$

Theorem 14 (Schur's Lemma for Representations). *We get this new version of Schur's lemma by translating into the language of representations.*

If V is an irreducible representation of a group G , then every entertwiner $T \in \text{End}_G(V)$ is invertible, that is $\text{End}_G(V)$ is a division ring.

Wedderburn's Theorem

Theorem 15 (Wedderburn's Theorem). *Let A be a finite-dimensional semisimple algebra over an algebraically closed field F . Then*

$$A \cong \bigoplus_{i=1}^r \text{Mat}_{d_i}(F)$$

for some positive integers d_i .

Remark 16. *In this course, we mostly use Wedderburn's theorem with $A = \mathbb{C}[G]$. By Maschke's theorem, $\mathbb{C}[G]$ is semisimple, so*

$$\mathbb{C}[G] \cong \bigoplus_{i=1}^r \text{Mat}_{d_i}(\mathbb{C}).$$

The integers d_i are exactly the dimensions of the irreducible representations of G .

Proposition 17. *If $V_1, \dots, V_r$ are the irreducible complex representations of G and $d_i = \dim V_i$, then*

$$\sum_{i=1}^r d_i^2 = |G|.$$

Proof. Since $\mathbb{C}[G] \cong \bigoplus_i \text{Mat}_{d_i}(\mathbb{C})$, taking dimensions over $\mathbb{C}$ gives

$$|G| = \dim_{\mathbb{C}} \mathbb{C}[G] = \sum_i \dim_{\mathbb{C}} \text{Mat}_{d_i}(\mathbb{C}) = \sum_i d_i^2.$$

$\square$

Remark 18. *This is one of the main tools for building character tables. Once we know some irreducible character degrees, the equation $\sum_i d_i^2 = |G|$ often forces the missing degrees.*

For the proof, see this video. The point for us is not the full proof, but the way it explains why semisimple algebras split into matrix algebras.

Characters and Character Tables

Characters as Class Functions

Definition 19 (Character). Let (π, V) be a finite-dimensional complex representation of G . The character of π is the function $\chi : G \rightarrow \mathbb{C}$ defined by

$$\chi(g) = \text{tr}(\pi(g)).$$

The degree of χ is $\chi(1) = \dim V$.

Proposition 20. Characters are constant on conjugacy classes.

Proof. Suppose $h = \lambda g \lambda^{-1}$. Then

$$\pi(h) = \pi(\lambda)\pi(g)\pi(\lambda)^{-1}.$$

So $\pi(g)$ and $\pi(h)$ are similar matrices. Similar matrices have the same trace, hence $\chi(g) = \chi(h)$. $\square$

Theorem 21. If A and B are square matrices of the same size, then

$$\text{tr}(AB) = \text{tr}(BA).$$

Proof. If $A = (a_{ij})$ and $B = (b_{ij})$, then

$$\text{tr}(AB) = \sum_i \sum_k a_{ik} b_{ki}$$

and

$$\text{tr}(BA) = \sum_i \sum_k b_{ik} a_{ki}.$$

Relabeling the indices shows these are equal. $\square$

Definition 22 (Class Function). A function $f : G \rightarrow \mathbb{C}$ is called a class function if $f(g) = f(hgh^{-1})$ for all $g, h \in G$.

Remark 23. If G has conjugacy classes $C_1, \dots, C_r$, then a class function is determined by its values on $C_1, \dots, C_r$. Hence the vector space of class functions has dimension r .

Remark 24. The character table has one column for each conjugacy class because characters are class functions. It has one row for each irreducible character.

The following theorem says that the number of irreducible complex characters equals the number of conjugacy classes, so the character table is square.

Theorem 25 (Number of Irreducible Characters).

Schur Orthogonality

Definition 26 (Inner Product of Class Functions). If $f_1, f_2 : G \longrightarrow \mathbb{C}$ are complex-valued functions, define

$$\langle f_1, f_2 \rangle = \frac{1}{|G|} \sum_{g \in G} f_1(g) \overline{f_2(g)}.$$

Theorem 27 (Schur Orthogonality). Let $(\pi_1, V_1), \dots, (\pi_r, V_r)$ be pairwise non-isomorphic irreducible complex representations of G , with characters $\chi_1, \dots, \chi_r$. Then

$$\langle \chi_i, \chi_j \rangle = \delta_{ij}.$$

Remark 28. This means irreducible characters are orthonormal. In particular, if χ is irreducible, then $\langle \chi, \chi \rangle = 1$. More generally, if $\chi = \sum_i m_i \chi_i$, then $\langle \chi, \chi_i \rangle = m_i$.

Corollary 29. A character χ is irreducible if and only if $\langle \chi, \chi \rangle = 1$.

Remark 30. This criterion is extremely useful in examples. When we construct a new character, we can often check irreducibility by computing its inner product with itself.

The Regular Representation

Definition 31 (Regular Representation). Let G be a finite group. The left regular representation is obtained from the action $G \curvearrowright G$ given by $g \cdot x = gx$. Linearizing this action gives a representation $\rho : G \longrightarrow GL(\mathbb{C}[G])$, where $\mathbb{C}[G]$ has basis $\{e_x : x \in G\}$ and $\rho(g)e_x = e_{gx}$.

Proposition 32. The character of the regular representation is

$$\chi_{\text{reg}}(g) = \begin{cases} |G|, & g = 1, \\ 0, & g \neq 1. \end{cases}$$

Proof. The matrix of $\rho(g)$ permutes the basis vectors e_x . The trace of a permutation matrix is the number of fixed basis vectors. Now $gx = x$ if and only if $g = 1$, so every basis vector is fixed by 1, and no basis vector is fixed by $g \neq 1$. $\square$

Proposition 33. If $V_1, \dots, V_r$ are the irreducible representations of G and $d_i = \dim V_i$, then

$$\mathbb{C}[G] \cong \bigoplus_{i=1}^r d_i V_i$$

as a representation of G .

Remark 34. The regular representation contains every irreducible representation, and the multiplicity of V_i is its dimension d_i .

This post gives a conceptual explanation of why orthogonality relations compute spaces of equivariant maps.

One-Dimensional Characters and the Commutator Subgroup

Definition 35 (Commutator Subgroup). The commutator subgroup of G is

$$G' = [G, G] = \langle aba^{-1}b^{-1} : a, b \in G \rangle.$$

Proposition 36. *The one-dimensional characters of G are exactly the one-dimensional characters of G/G' .*

Proof. A one-dimensional character is a homomorphism $\chi : G \longrightarrow \mathbb{C}^\times$. Since $\mathbb{C}^\times$ is abelian, every commutator must map to 1. Therefore $G' \subseteq \ker \chi$, so χ factors through G/G' . $\square$

Corollary 37. For $g \in G$,

$$g \in G' \iff \chi(g) = 1$$

for every degree 1 character χ .

Remark 38. This gives a way to read off G' from the character table. Look at the rows corresponding to degree 1 characters. The elements on which all of those characters take value 1 are exactly the elements of the commutator subgroup.

Abelian Groups

Proposition 39. *If G is abelian, then every irreducible complex representation of G is one-dimensional.*

Proof. Since G is abelian, every conjugacy class has size 1, so there are $|G|$ conjugacy classes. Hence there are $|G|$ irreducible characters. If their degrees are $d_1, \dots, d_{|G|}$, then

$$\sum_i d_i^2 = |G|.$$

Since each $d_i \geq 1$, all d_i must equal 1. $\square$

Example 40. Let $G = \mathbb{Z}_2 = \langle x \rangle$. Then the two irreducible characters are

	1	x
χ_1	1	1
χ_2	1	-1

Remark 41. For a cyclic group $C_n = \langle x \rangle$, the irreducible complex characters are given by $\chi_k(x) = \zeta_n^k$, where $\zeta_n = e^{2\pi i/n}$ and $0 \leq k \leq n-1$.

Remark 42. The entries in the row of a character χ are values of χ on conjugacy class representatives. When computing inner products from a character table, remember to weight by conjugacy class sizes:

$$\langle \chi, \psi \rangle = \frac{1}{|G|} \sum_{\text{classes } C} |C| \chi(C) \overline{\psi(C)}.$$

Roots of Unity and the Triangle Inequality

Proposition 43. *If $g \in G$ has finite order and π is a complex representation of G , then the eigenvalues of $\pi(g)$ are roots of unity.*

Proof. If $g^n = 1$, then $\pi(g)^n = I$. Hence each eigenvalue ε of $\pi(g)$ satisfies $\varepsilon^n = 1$. $\square$

Corollary 44. *For every character χ ,*

$$\chi(g^{-1}) = \overline{\chi(g)}.$$

Proof. The eigenvalues of $\pi(g^{-1})$ are the inverses of the eigenvalues of $\pi(g)$. Since these eigenvalues are roots of unity, inversion is the same as complex conjugation. $\square$

Remark 45. *The triangle inequality is used constantly. If $\varepsilon_1, \dots, \varepsilon_d$ are roots of unity, then*

$$|\varepsilon_1 + \dots + \varepsilon_d| \leq |\varepsilon_1| + \dots + |\varepsilon_d| = d.$$

Equality occurs exactly when all ε_i point in the same direction.

Detecting Kernels, Centers, and Normal Subgroups

Theorem 46. *Let χ be the character of a representation $\pi : G \longrightarrow GL(V)$. Then*

$$g \in \ker \pi \iff \chi(g) = \chi(1).$$

Proof. If $g \in \ker \pi$, then $\pi(g) = I_V$, so $\chi(g) = \text{tr}(I_V) = \dim V = \chi(1)$.

Conversely, suppose $\chi(g) = \chi(1) = d$. The eigenvalues of $\pi(g)$ are roots of unity $\varepsilon_1, \dots, \varepsilon_d$, so

$$d = \chi(g) = \varepsilon_1 + \dots + \varepsilon_d.$$

Since each $|\varepsilon_i| = 1$, equality in the triangle inequality forces $\varepsilon_1 = \dots = \varepsilon_d = 1$. Hence $\pi(g) = I_V$, so $g \in \ker \pi$. $\square$

Corollary 47. *If χ is a character, then*

$$\ker \chi = \{g \in G : \chi(g) = \chi(1)\}$$

is a normal subgroup.

Proposition 48. *Let $g \in G$. Then $g \in Z(G)$ if and only if $|\chi(g)| = \chi(1)$ for every irreducible character χ of G .*

Proof. If $g \in Z(G)$ and (π, V) is irreducible, then $\pi(g)$ commutes with every $\pi(h)$. By Schur's lemma, $\pi(g) = \lambda I$ for some $\lambda \in \mathbb{C}$. Since g has finite order, λ is a root of unity. Therefore

$$|\chi(g)| = |\text{tr}(\lambda I)| = |\lambda \dim V| = \dim V = \chi(1).$$

Conversely, suppose $|\chi(g)| = \chi(1)$ for every irreducible character χ . For every irreducible representation π , the triangle inequality argument shows that $\pi(g)$ is a scalar matrix. Hence g acts as a scalar in every irreducible representation. Since the regular representation contains every irreducible representation, left multiplication by g commutes with the regular action of G . Therefore g is central. $\square$

Remark 49. Character tables can detect normal subgroups because kernels of characters are normal subgroups, and intersections of such kernels are normal subgroups. Since characters are constant on conjugacy classes, these normal subgroups appear as unions of conjugacy classes.

Permutation Characters

Definition 50 (Permutation Character). Suppose G acts on a finite set X . Let $V = \mathbb{C}[X]$ be the vector space with basis $\{e_x : x \in X\}$ and define $ge_x = e_{gx}$. The character of this representation is called the permutation character of the action.

Proposition 51. If θ is the permutation character of the action $G \curvearrowright X$, then

$$\theta(g) = \#\{x \in X : gx = x\}.$$

Proof. The matrix of g permutes the basis vectors e_x . The trace of a permutation matrix equals the number of basis vectors fixed by the permutation. $\square$

Remark 52. If G acts transitively on X , then the line spanned by $\sum_{x \in X} e_x$ is always a trivial subrepresentation. Often the complementary representation has character $\theta - 1$, but it is not always irreducible. One checks irreducibility using $\langle \theta - 1, \theta - 1 \rangle = 1$.

Tensor Products of Representations

Proposition 53. If (π, V) and (θ, W) are representations of G , then $(\pi \otimes \theta, V \otimes W)$ is a representation of G , where

$$(\pi \otimes \theta)(g) = \pi(g) \otimes \theta(g).$$

If χ and ψ are the corresponding characters, then the character of $V \otimes W$ is $\chi\psi$.

Proof. First,

$$(\pi(g) \otimes \theta(g))(\pi(h) \otimes \theta(h)) = \pi(gh) \otimes \theta(gh),$$

so this is a representation. If $A \in \text{End}(V)$ and $B \in \text{End}(W)$, then

$$\text{tr}(A \otimes B) = \text{tr}(A) \text{tr}(B).$$

Applying this to $A = \pi(g)$ and $B = \theta(g)$ gives

$$\chi_{V \otimes W}(g) = \chi_V(g) \chi_W(g).$$

$\square$

Remark 54. The product of two irreducible characters is not necessarily irreducible. However, if one factor is a degree 1 character, then multiplying by it preserves irreducibility.

Example: The Character Table of D_8

Example 55. Let

$$D_8 = \langle r, s \mid r^4 = s^2 = 1, srs^{-1} = r^{-1} \rangle.$$

The commutator subgroup is $D'_8 = \langle r^2 \rangle$, so

$$D_8/D'_8 \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

Therefore D_8 has four one-dimensional characters.

The conjugacy classes are represented by $1, r, r^2, s, rs$, so there are five irreducible characters. Since

$$1^2 + 1^2 + 1^2 + 1^2 + d^2 = 8,$$

the remaining character has degree 2.

The character table is

	1	r	r^2	s	rs
χ_1	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	1	-1	1	1	-1
χ_4	1	-1	1	-1	1
χ_5	2	0	-2	0	0

Remark 56. The degree 2 character comes from the standard geometric representation of D_8 as the symmetries of a square. Rotations and reflections become 2×2 matrices, and the character values are their traces.

Example: The Character Table of D_{12}

Example 57. Let

$$D_{12} = \langle r, s \mid r^6 = s^2 = 1, srs = r^{-1} \rangle.$$

The conjugacy class representatives are

$$1, r, r^2, r^3, s, rs.$$

There are six conjugacy classes, so there are six irreducible characters. We have $G' = \langle r^2 \rangle$, so $|G/G'| = 4$. Therefore there are four degree 1 characters.

The degree equation gives

$$1^2 + 1^2 + 1^2 + 1^2 + d_5^2 + d_6^2 = 12,$$

so $d_5 = d_6 = 2$.

The character table is

	1	r	r^2	r^3	s	rs
χ_1	1	1	1	1	1	1
χ_2	1	1	1	1	-1	-1
χ_3	1	-1	1	-1	1	-1
χ_4	1	-1	1	-1	-1	1
χ_5	2	1	-1	-2	0	0
χ_6	2	-1	-1	2	0	0

Keith Conrad's notes on dihedral groups are good background for conjugacy classes and presentations. These notes compute the characters of dihedral groups directly.

Remark 58. To construct χ_5 , use

$$R = \begin{pmatrix} \cos(2\pi/6) & -\sin(2\pi/6) \\ \sin(2\pi/6) & \cos(2\pi/6) \end{pmatrix}, \quad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then $R^6 = I$, $S^2 = I$, and $SRS = R^{-1}$. The character is obtained by taking traces. The final degree 2 character can be found by multiplying χ_5 by a suitable degree 1 character.

Example: The Character Table of A_4

Example 59. The group A_4 is not abelian. It has order 12. Its conjugacy classes have representatives

$$1, \quad (12)(34), \quad (123), \quad (132).$$

Hence A_4 has four irreducible characters.

The Klein four subgroup

$$V = \{1, (12)(34), (13)(24), (14)(23)\}$$

is normal in A_4 , and in fact

$$A_4' = V.$$

Therefore

$$A_4/V \cong C_3.$$

This gives three degree 1 characters. Let $\rho = e^{2\pi i/3}$. The character table of A_4/V is

A_4/V	1	t	t^2
χ_1	1	1	1
χ_2	1	ρ	ρ^2
χ_3	1	ρ^2	ρ

where $t = \overline{(123)}$.

Pulling these characters back to A_4 , and using the degree equation

$$1^2 + 1^2 + 1^2 + d^2 = 12,$$

we get $d = 3$. The character table is

A_4	1	(12)(34)	(123)	(132)
χ_1	1	1	1	1
χ_2	1	1	ρ	ρ^2
χ_3	1	1	ρ^2	ρ
χ_4	3	-1	0	0

Remark 60. The character χ_4 comes from the action of A_4 on $X = \{1, 2, 3, 4\}$. The permutation character θ counts fixed points. Since the line spanned by $e_1 + e_2 + e_3 + e_4$ is trivial, the character $\theta - 1$ has degree 3. In this case, $\theta - 1$ is irreducible.

Remark 61. The two types of 3-cycles split into two conjugacy classes in A_4 , even though they are conjugate in S_4 . This is why (123) and (132) appear as different columns.

A useful computational reference for small character tables is GroupNames.

Central Idempotents in $\mathbb{C}[G]$

Proposition 62. Let χ_i be an irreducible character of degree d_i . Define

$$e_i = \frac{d_i}{|G|} \sum_{g \in G} \chi_i(g^{-1})g.$$

Then e_i is a central idempotent in $\mathbb{C}[G]$.

Remark 63. Central means $e_i x = x e_i$ for every $x \in \mathbb{C}[G]$. Idempotent means $e_i^2 = e_i$. These elements project $\mathbb{C}[G]$ onto the isotypic components corresponding to the irreducible representations.

Remark 64. If $i \neq j$, then $e_i e_j = 0$. This is the group algebra version of Schur orthogonality.

Checklist for Small Character Tables

Scholium 65. To compute character tables for small groups, use the following checklist.

1. Find the conjugacy classes and their sizes.
2. Count the conjugacy classes to know the number of irreducible characters.
3. Put the principal character first.
4. Compute G' and G/G' .
5. Use G/G' to fill in all degree 1 characters.
6. Use $\sum_i d_i^2 = |G|$ to find the remaining degrees.
7. Look for geometric representations, especially for dihedral groups.
8. Look for permutation representations, especially when the group acts naturally on a small set.
9. Use tensor products with degree 1 characters to produce more rows.
10. Check row orthogonality.
11. Check column orthogonality if needed.
12. Use kernels of characters to identify normal subgroups.

Induced Representations

Proposition 66. Let (π, V) and (θ, W) be finite-dimensional complex representations of a finite group G . Let χ and ψ be their characters. Then

$$(\chi, \psi)_G = \dim \operatorname{Hom}_{\mathbb{C}[G]}(V, W).$$

Proof. Since G is finite and we are working over $\mathbb{C}$, every representation is completely reducible. Write

$$V \cong \bigoplus_i m_i U_i, \quad W \cong \bigoplus_i n_i U_i,$$

where the U_i are pairwise non-isomorphic irreducible representations of G . If χ_i is the character of U_i , then

$$\chi = \sum_i m_i \chi_i, \quad \psi = \sum_i n_i \chi_i.$$

By Schur orthogonality,

$$(\chi_i, \chi_j)_G = \begin{cases} 1 & \text{if } U_i \cong U_j, \\ 0 & \text{otherwise.} \end{cases}$$

Therefore

$$(\chi, \psi)_G = \left(\sum_i m_i \chi_i, \sum_j n_j \chi_j \right)_G = \sum_i m_i n_i.$$

On the other hand,

$$\text{Hom}_{\mathbb{C}[G]}(V, W) \cong \bigoplus_{i,j} \text{Hom}_{\mathbb{C}[G]}(m_i U_i, n_j U_j).$$

By Schur's lemma, $\text{Hom}_{\mathbb{C}[G]}(U_i, U_j) = 0$ if $i \neq j$, and $\text{Hom}_{\mathbb{C}[G]}(U_i, U_i) \cong \mathbb{C}$. Hence

$$\dim \text{Hom}_{\mathbb{C}[G]}(V, W) = \sum_i m_i n_i.$$

So

$$(\chi, \psi)_G = \dim \text{Hom}_{\mathbb{C}[G]}(V, W).$$

□

Remark 67. The character inner product detects overlap between representations. If V and W are irreducible, then it is 1 when they are the same representation and 0 otherwise. For general representations, it counts the number of matching irreducible pieces, with multiplicity.

Theorem 68 (Frobenius Reciprocity). Let $H \leq G$. If χ is a character of H and ψ is a character of G , then

$$(\text{Ind}_H^G \chi, \psi)_G = (\chi, \text{Res}_H^G \psi)_H.$$

Remark 69. The restriction symbol is important. The left side compares class functions on G , while the right side compares class functions on H .

Definition 70 (Induced Character). Let χ be a character of H . Define the extension-by-zero function $\chi^0 : G \rightarrow \mathbb{C}$ by

$$\chi^0(g) = \begin{cases} \chi(g) & \text{if } g \in H, \\ 0 & \text{if } g \notin H. \end{cases}$$

Induction is adjoint to restriction. For more digging on this, see this.

The induced character $\text{Ind}_H^G \chi$ is the class function on G given by

$$(\text{Ind}_H^G \chi)(x) = \frac{1}{|H|} \sum_{g \in G} \chi^0(g^{-1}xg).$$

More on this here.

Proposition 71. *The class function*

$$\chi^G(x) = \frac{1}{|H|} \sum_{g \in G} \chi^0(g^{-1}xg)$$

satisfies Frobenius reciprocity:

$$(\chi^G, \psi)_G = (\chi, \text{Res}_H^G \psi)_H.$$

Therefore χ^G is the character of the induced representation $\text{Ind}_H^G \chi$.

Proof. We compute:

$$\begin{aligned} (\chi^G, \psi)_G &= \frac{1}{|G|} \sum_{x \in G} \chi^G(x) \overline{\psi(x)} \\ &= \frac{1}{|G|} \sum_{x \in G} \left(\frac{1}{|H|} \sum_{g \in G} \chi^0(g^{-1}xg) \right) \overline{\psi(x)} \\ &= \frac{1}{|G||H|} \sum_{g \in G} \sum_{x \in G} \chi^0(g^{-1}xg) \overline{\psi(x)}. \end{aligned}$$

Set $y = g^{-1}xg$, so $x = gyg^{-1}$. Since ψ is a character of G , it is constant on conjugacy classes, so $\psi(gyg^{-1}) = \psi(y)$. Therefore

$$\begin{aligned} (\chi^G, \psi)_G &= \frac{1}{|G||H|} \sum_{g \in G} \sum_{y \in G} \chi^0(y) \overline{\psi(y)} \\ &= \frac{1}{|H|} \sum_{y \in G} \chi^0(y) \overline{\psi(y)} \\ &= \frac{1}{|H|} \sum_{y \in H} \chi(y) \overline{\psi(y)} \\ &= (\chi, \text{Res}_H^G \psi)_H. \end{aligned}$$

□

Remark 72. *The induced character formula says that to evaluate $\text{Ind}_H^G \chi$ at $x \in G$, we look at all conjugates of x which land inside H , evaluate χ there, and average correctly. If no conjugate of x lies in H , then the induced character is 0 at x .*

Theorem 73 (Brauer Induction). *Let G be a finite group. Every complex character χ of G can be written as an integer linear combination*

$$\chi = \sum_i n_i \text{Ind}_{H_i}^G(\psi_i), \quad n_i \in \mathbb{Z},$$

where each $H_i \leq G$ and each ψ_i is a 1-dimensional character of H_i .

For a matrix-level picture of induction, see The Unapologetic Mathematician, *Induced Matrix Representations*: <https://unapologetic.wordpress.com/2010/11/25/induced-matrix-representations/>. For Frobenius reciprocity in the form $\text{Hom}_G(\text{Ind}_H^G V, W) \cong \text{Hom}_H(V, \text{Res}_H^G W)$, see The Unapologetic Mathematician, *Real Frobenius Reciprocity*: <https://unapologetic.wordpress.com/2010/12/03/real-frobenius-reciprocity/>.

Remark 74. The significance is that induction from 1-dimensional characters is powerful enough to build every character, as long as we allow integer linear combinations.

Theorem 75. If G is a finite p -group, then every irreducible complex representation of G is induced from a 1-dimensional representation of some subgroup.

Remark 76. For p -groups, this is stronger than Brauer induction. We do not merely get an integer linear combination of induced characters. Each irreducible representation itself is induced from a 1-dimensional representation of some subgroup.

Example 77 (D_{12}). Let

$$D_{12} = \langle r, s \mid r^6 = s^2 = 1, srs^{-1} = r^{-1} \rangle.$$

Here $|D_{12}| = 12$, and the subgroup $H = \langle r \rangle \cong C_6$ has index 2.

For $k = 0, 1, \dots, 5$, define a 1-dimensional character of H by

$$\chi_k(r) = \zeta^k, \quad \zeta = e^{2\pi i/6}.$$

We induce χ_k from H to D_{12} :

$$\text{Ind}_H^{D_{12}} \chi_k.$$

Since H is normal in D_{12} and $srs^{-1} = r^{-1}$, for $x = r^m$ we get

$$(\text{Ind}_H^{D_{12}} \chi_k)(r^m) = \chi_k(r^m) + \chi_k(r^{-m}) = \zeta^{km} + \zeta^{-km} = 2 \cos\left(\frac{2\pi km}{6}\right).$$

For a reflection $x = sr^m$, no conjugate of x lies in H , so

$$(\text{Ind}_H^{D_{12}} \chi_k)(sr^m) = 0.$$

Therefore

$$(\text{Ind}_H^{D_{12}} \chi_k)(r^m) = 2 \cos\left(\frac{2\pi km}{6}\right), \quad (\text{Ind}_H^{D_{12}} \chi_k)(sr^m) = 0.$$

The cases $k = 1$ and $k = 2$ give the two irreducible 2-dimensional representations of D_{12} . The cases $k = 0$ and $k = 3$ are reducible, and $k = 4, 5$ give the same representations as $k = 2, 1$ respectively, because χ_k and χ_{-k} induce to the same character.

The induced representation has dimension

$$[D_{12} : H] \dim \chi_k = 2.$$

So inducing a 1-dimensional character of the rotation subgroup naturally produces 2-dimensional representations of the dihedral group.

Frobenius' Theorem on Frobenius Groups

We now use character theory to prove a theorem whose statement is purely group-theoretic. This is one of the standard examples where representations say something directly about the internal structure of a finite group.

Definition 78. Let G be a finite group and let $H \leq G$ be a nontrivial proper subgroup. We say that H is a **Frobenius complement** if $H \cap gHg^{-1} = \{1\}$ for every $g \notin H$. In this case, G is called a **Frobenius group** with complement H .

Equivalently, the nontrivial conjugates of H are as disjoint as possible. The theorem is that all elements not lying in one of these conjugates, together with the identity, secretly form a normal subgroup.

Definition 79. Let G be a Frobenius group with complement H . Define

$$K = \left(G \setminus \bigcup_{g \in G} g(H \setminus \{1\})g^{-1} \right) \cup \{1\}.$$

The set K is called the **Frobenius kernel**.

Example 80. Let $G = S_3$ and $H = \langle (12) \rangle$. Then the conjugates of H contain the three transpositions. Hence $K = \{1, (123), (132)\} = A_3$. So in this case $S_3 = A_3 \rtimes H$.

Example 81. Let $\mathbb{F}_q$ be a finite field and let G be the group of affine maps $x \mapsto ax + b$ where $a \in \mathbb{F}_q^\times$ and $b \in \mathbb{F}_q$. The subgroup H of maps $x \mapsto ax$ is a Frobenius complement. The kernel is the subgroup of translations $x \mapsto x + b$.

Theorem 82 (Frobenius). Let G be a Frobenius group with complement H , and let K be the Frobenius kernel. Then $K \triangleleft G$ and

$$G = K \rtimes H.$$

Proof. We first record the character-theoretic input. Let χ be a character of H with $\chi(1) = 0$. Extend it to a function $\chi^\sharp$ on G by setting

$$\chi^\sharp(x) = \begin{cases} \chi(h) & \text{if } x \text{ is conjugate in } G \text{ to } h \in H, \\ 0 & \text{if no conjugate of } x \text{ lies in } H. \end{cases}$$

This is well-defined because if $h_1, h_2 \in H \setminus \{1\}$ are conjugate in G , then they are already conjugate in H . Indeed, if $h_2 = gh_1g^{-1}$, then $H \cap gHg^{-1}$ contains the nonidentity element h_2 , so $g \in H$.

Now compare this with the induced character formula. Since $\chi(1) = 0$, for $x \in H \setminus \{1\}$ the only conjugates contributing to $\text{Ind}_H^G \chi(x)$ come from H itself, and we get $\text{Ind}_H^G \chi(x) = \chi(x)$. If x is not conjugate into H , the induced character is 0. Also $\text{Ind}_H^G \chi(1) = [G : H]\chi(1) = 0$. Hence $\chi^\sharp = \text{Ind}_H^G \chi$, so $\chi^\sharp$ is a generalized character of G .

Apply this to $\chi = \theta - \theta(1)1_H$, where θ ranges over the irreducible characters of H . The resulting generalized characters on G vanish on K and agree with $\theta - \theta(1)$ on the nonidentity elements of H and their conjugates.

The orthogonality relations now force the elements of K to behave like a subgroup. More precisely, one uses the column orthogonality relations for G , together with the above induced generalized characters, to show that if $x, y \in K$, then $xy^{-1} \in K$. Since $1 \in K$ and K is closed under inverses by definition, this proves that K is a subgroup.

Tao's post on Frobenius and Suzuki gives good context for why this theorem is a serious group-theoretic application of characters: <https://terrytao.wordpress.com/2013/04/12/the-theorems-of-frobenius-and-suzuki-on-fini>

Finally, K is a union of conjugacy classes by definition, so $K \triangleleft G$. Also $K \cap H = \{1\}$, and the counting gives

$$|K| = |G| - \left| \bigcup_{g \in G} g(H \setminus \{1\})g^{-1} \right| + 1 = |G| - [G : H](|H| - 1) = [G : H].$$

Therefore $|KH| = |K||H| = |G|$, so $G = KH$. Since $K \triangleleft G$ and $K \cap H = \{1\}$, this is the semidirect product $G = K \rtimes H$. $\square$

Remark 83. *The striking point is that normality of K is not clear from the definition. The set K is visibly stable under conjugation, but it is not visibly closed under multiplication. Character theory is what proves the closure.*

For another viewpoint, Tao gives a Fourier-analytic proof of Frobenius' theorem here: <https://terrytao.wordpress.com/2013/05/24/a-fourier-analytic-proof-of-frobenius-theorem/>

Finitely Generated Modules over a PID

Let R be a PID. The two main examples to keep in mind are:

1. $R = \mathbb{Z}$. Then R -modules are abelian groups.
2. $R = F[X]$. If V is a finite-dimensional F -vector space and $T : V \rightarrow V$ is linear, then V becomes an $F[X]$ -module by

$$f(X) \cdot v = f(T)v.$$

This is the module-theoretic source of rational canonical form.

Torsion and Length

Definition 84. Let R be an integral domain and let M be an R -module. The torsion submodule is

$$T(M) = \text{Tor}(M) = \{m \in M : \exists a \neq 0 \in R \text{ such that } am = 0\}.$$

We say M is **torsion** if $T(M) = M$, and **torsion-free** if $T(M) = 0$.

Remark 85. *The integral domain condition is needed because if $x, y \in T(M)$, say $ax = 0$ and $by = 0$ with $a, b \neq 0$, then*

$$ab(x + y) = 0.$$

Since R is an integral domain, $ab \neq 0$, so $x + y \in T(M)$.

Definition 86. A finite-length module M has a chain

$$0 = M_0 \subset M_1 \subset \cdots \subset M_k = M$$

where each quotient M_i/M_{i-1} is simple. The **length** of M is

$$l(M) = k.$$

Over a PID, the simple modules are the modules

$$R/(p),$$

where $p \in R$ is irreducible.

Example 87. For $R = \mathbb{Z}$,

$$l(\mathbb{Z}/12\mathbb{Z}) = 3,$$

since $12 = 2^2 \cdot 3$. More generally,

$$l(\mathbb{Z}/p^n\mathbb{Z}) = n.$$

Definition 88. If M is p -torsion and $x \in M$, then x has **order** p^k if

$$p^k x = 0, \quad p^{k-1} x \neq 0.$$

An element of maximal order is an element whose order is as large as possible among all elements of M .

Free Part and Torsion Part

Theorem 89. Let R be an integral domain. Then $M/T(M)$ is torsion-free.

Proof. Suppose $a\bar{m} = 0$ in $M/T(M)$ for some $a \neq 0$. Then $am \in T(M)$, so there exists $b \neq 0$ with

$$b(am) = 0.$$

Since R is an integral domain, $ba \neq 0$, so $m \in T(M)$. Hence $\bar{m} = 0$. □

Theorem 90. If R is a PID and M is finitely generated and torsion-free, then

$$M \cong R^r$$

for some r .

Remark 91. This is one of the key places where the PID hypothesis matters. Over a general integral domain, finitely generated torsion-free modules need not be free.

Theorem 92 (Free plus torsion decomposition). If R is a PID and M is finitely generated, then

$$M \cong R^r \oplus T(M).$$

Proof. Since $M/T(M)$ is finitely generated and torsion-free, it is free:

$$M/T(M) \cong R^r.$$

Choose lifts $m_1, \dots, m_r \in M$ of a basis of $M/T(M)$, and let

$$F = Rm_1 + \dots + Rm_r.$$

Then $F \cong R^r$, and the quotient map gives an isomorphism

$$F \cong M/T(M).$$

Therefore every $m \in M$ can be written as $m = f + t$ with $f \in F$ and $t \in T(M)$. Also

$$F \cap T(M) = 0,$$

because F is torsion-free. Hence

$$M = F \oplus T(M) \cong R^r \oplus T(M).$$

□

Thus the classification reduces to the case where M is finitely generated and torsion.

Chinese Remainder Theorem

Theorem 93 (Chinese Remainder Theorem). *If $(a, b) = 1$ in a PID R , then*

$$R/(ab) \cong R/(a) \oplus R/(b).$$

Proof. Define

$$\phi : R \rightarrow R/(a) \oplus R/(b), \quad \phi(x) = (x + (a), x + (b)).$$

Since $(a, b) = 1$, choose $u, v \in R$ such that

$$ua + vb = 1.$$

For any $m \in R$,

$$\phi(mvb) = (m + (a), 0), \quad \phi(mua) = (0, m + (b)).$$

Hence ϕ is surjective. Also

$$\ker \phi = (a) \cap (b) = (ab),$$

since a and b are relatively prime. By the first isomorphism theorem,

$$R/(ab) \cong R/(a) \oplus R/(b).$$

□

Example 94.

$$\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}.$$

This is why decompositions into cyclic modules are not unique unless we impose a normal form.

Primary Decomposition

Fix an irreducible element $p \in R$. For a torsion module M , define

$$M(p) = \{m \in M : \exists n \geq 1 \text{ such that } p^n m = 0\}.$$

Example 95. If $R = \mathbb{Z}$ and M is a finite abelian group, then $M(p)$ is the usual p -primary component, analogous to a Sylow subgroup.

Theorem 96 (Primary decomposition). *If M is a finitely generated torsion module over a PID, then*

$$M = \bigoplus_p M(p),$$

where only finitely many summands are nonzero.

Proof. Since M is finitely generated and torsion, choose generators $m_1, \dots, m_s$. For each i , choose $a_i \neq 0$ with $a_i m_i = 0$. Then

$$aM = 0$$

for $a = a_1 \cdots a_s$. Therefore only irreducibles dividing a can appear.

Fix an irreducible p and write

$$a = p^N b, \quad (p, b) = 1.$$

Let

$$Q = \{x \in M : \exists c \neq 0 \text{ with } cx = 0 \text{ and } p \nmid c\}.$$

We claim

$$M = M(p) \oplus Q.$$

First, if $x \in M(p) \cap Q$, then $p^r x = 0$ and $cx = 0$ for some r and some c with $p \nmid c$. Since $(p^r, c) = 1$, choose $\alpha, \beta \in R$ such that

$$\alpha p^r + \beta c = 1.$$

Then

$$x = (\alpha p^r + \beta c)x = 0.$$

Hence $M(p) \cap Q = 0$.

Now let $x \in M$. Since $ax = 0$ and $a = p^N b$ with $(p^N, b) = 1$, choose $u, v \in R$ such that

$$up^N + vb = 1.$$

Then

$$x = up^N x + vb x.$$

The first term lies in Q because it is killed by b , and the second lies in $M(p)$ because it is killed by p^N . Hence

$$M = M(p) + Q.$$

Therefore

$$M = M(p) \oplus Q.$$

Repeating this splitting for each irreducible divisor of a gives

$$M = \bigoplus_{p|a} M(p).$$

□

Classification of the p -Primary Part

Theorem 97. *Let M be a finitely generated p -torsion module over a PID. Then there exist integers*

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k \geq 1$$

such that

$$M \cong \bigoplus_{i=1}^k R/(p^{\lambda_i}).$$

Proof. We induct on the length $l(M)$. The case $M = 0$ is trivial.

Choose $m \in M$ of maximal order, say

$$p^N m = 0, \quad p^{N-1} m \neq 0.$$

Then

$$Rm \cong R/(p^N).$$

Consider the quotient

$$\overline{M} = M/Rm.$$

Since $l(\overline{M}) < l(M)$, induction gives

$$\overline{M} \cong \bigoplus_{i=2}^k R/(p^{\lambda_i}).$$

Choose generators $\bar{x}_2, \dots, \bar{x}_k$ of these cyclic summands, where $\bar{x}_i$ has order p^{λ_i} . By the lifting lemma below, choose representatives $x_i \in M$ with the same orders. Let

$$N = Rx_2 + \dots + Rx_k.$$

The quotient map $M \rightarrow M/Rm$ restricts to an isomorphism

$$N \cong \overline{M}.$$

Therefore $M = Rm + N$ and $Rm \cap N = 0$. Hence

$$M = Rm \oplus N \cong R/(p^N) \oplus \bigoplus_{i=2}^k R/(p^{\lambda_i}).$$

□

Lemma 98 (Lifting lemma). *Let $m \in M$ have maximal order p^N . If $\bar{x} \in M/Rm$ has order p^k , then $\bar{x}$ has a representative $x \in M$ of order p^k .*

Proof. Let x_0 be any representative of $\bar{x}$. Since $\bar{x}$ has order p^k ,

$$p^k x_0 \in Rm.$$

So

$$p^k x_0 = tm$$

for some $t \in R$. Since m has maximal order p^N , all of M is killed by p^N , so

$$0 = p^N x_0 = p^{N-k} tm.$$

Since $\text{ann}(m) = (p^N)$, this implies

$$p^N \mid p^{N-k} t,$$

so

$$p^k \mid t.$$

Write $t = p^k u$. Define

$$x = x_0 - um.$$

Then x is still a representative of $\bar{x}$, and

$$p^k x = p^k x_0 - p^k um = tm - tm = 0.$$

Since $\bar{x}$ has order exactly p^k in the quotient, x cannot have smaller order. Thus x has order p^k . $\square$

Structure Theorem

Theorem 99 (Structure theorem, elementary divisor form). *Let R be a PID and let M be a finitely generated R -module. Then*

$$M \cong R^r \oplus \bigoplus_p \bigoplus_i R/(p^{\lambda_{p,i}}),$$

where p ranges over irreducibles of R , only finitely many summands occur, and

$$\lambda_{p,1} \geq \lambda_{p,2} \geq \dots$$

for each fixed p .

The factors

$$p^{\lambda_{p,i}}$$

are called the **elementary divisors** of M .

Theorem 100 (Structure theorem, invariant factor form). *Let R be a PID and let M be a finitely generated R -module. Then*

$$M \cong R^r \oplus R/(a_1) \oplus \dots \oplus R/(a_t),$$

where

$$a_1 \mid a_2 \mid \dots \mid a_t.$$

The elements a_i are called the **invariant factors**.

Elementary Divisors and Invariant Factors

The elementary divisors and invariant factors are two ways of recording the same torsion module.

Starting from elementary divisors, group them by irreducible p :

$$p: \quad p^{e_{p,1}}, p^{e_{p,2}}, \dots, p^{e_{p,r_p}},$$

with

$$e_{p,1} \leq e_{p,2} \leq \dots \leq e_{p,r_p}.$$

Pad shorter rows on the left using $p^0 = 1$. Then multiply down the columns:

$$a_j = \prod_p p^{e_{p,j}}.$$

The column construction guarantees

$$a_1 \mid a_2 \mid \cdots \mid a_t.$$

By the Chinese Remainder Theorem,

$$R/(a_j) \cong \bigoplus_p R/(p^{e_{p,j}}),$$

because the different prime powers are relatively prime.

Example 101. Suppose

$$M \cong \mathbb{Z}/8 \oplus \mathbb{Z}/4 \oplus \mathbb{Z}/9 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/5.$$

The elementary divisors are

$$2^3, 2^2, 3^2, 3, 5.$$

Group by prime and order increasingly:

$$2: \quad 2^2, 2^3, \quad 3: \quad 3^1, 3^2, \quad 5: \quad 5^1.$$

Pad on the left:

$$2: \quad 2^2, 2^3, \quad 3: \quad 3^1, 3^2, \quad 5: \quad 1, 5.$$

Multiply down columns:

$$a_1 = 2^2 \cdot 3 \cdot 1 = 12, \quad a_2 = 2^3 \cdot 3^2 \cdot 5 = 360.$$

Thus

$$M \cong \mathbb{Z}/12 \oplus \mathbb{Z}/360,$$

and indeed

$$12 \mid 360.$$

Conversely, starting from invariant factors, factor each a_j into prime powers and use CRT:

$$R/(a_j) \cong \bigoplus_{p^e \parallel a_j} R/(p^e).$$

Uniqueness

The rank r is unique because

$$r = \dim_K(M \otimes_R K),$$

where $K = \text{Frac}(R)$.

For the torsion part, fix an irreducible p . If

$$M(p) \cong \bigoplus_i R/(p^{\lambda_i}),$$

then the dimensions

$$d_k = \dim_{R/(p)} \left(p^{k-1}M(p)/p^kM(p) \right)$$

satisfy

$$d_k = \#\{i : \lambda_i \geq k\}.$$

These numbers are intrinsic to M , so they determine the multiset of exponents $\{\lambda_i\}$. Hence the elementary divisors are unique up to ordering and multiplication by units. The invariant factors are then unique because they are obtained canonically from the elementary divisors by the column construction above.

Finitely Generated Abelian Groups

Taking $R = \mathbb{Z}$, every finitely generated abelian group has the form

$$A \cong \mathbb{Z}^r \oplus \mathbb{Z}/(a_1) \oplus \cdots \oplus \mathbb{Z}/(a_t), \quad a_1 \mid a_2 \mid \cdots \mid a_t.$$

Equivalently, its torsion part decomposes into prime-power cyclic groups:

$$T(A) \cong \bigoplus_p \bigoplus_i \mathbb{Z}/(p^{\lambda_{p,i}}).$$

Rational Canonical Form

Let V be a finite-dimensional vector space over F , and let $T : V \rightarrow V$ be linear. Make V into an $F[X]$ -module by

$$f(X) \cdot v = f(T)v.$$

Since V is finite-dimensional, it is a torsion $F[X]$ -module. Indeed, the minimal polynomial $\mu_T(X)$ satisfies

$$\mu_T(T)v = 0$$

for every $v \in V$.

Since $F[X]$ is a PID, the structure theorem gives

$$V \cong F[X]/(f_1) \oplus \cdots \oplus F[X]/(f_t), \quad f_1 \mid f_2 \mid \cdots \mid f_t.$$

The polynomials f_i are the invariant factors of T . The largest invariant factor f_t is the minimal polynomial of T , and

$$\prod_i f_i$$

is the characteristic polynomial of T .

If

$$f(X) = X^n + a_{n-1}X^{n-1} + \cdots + a_1X + a_0,$$

then $F[X]/(f)$ has basis

$$\bar{1}, \bar{X}, \dots, \bar{X}^{n-1}.$$

Multiplication by X sends

$$\bar{1} \mapsto \bar{X} \mapsto \cdots \mapsto \bar{X}^{n-1}$$

and

$$\overline{X}^n = -a_0\overline{1} - a_1\overline{X} - \cdots - a_{n-1}\overline{X}^{n-1}.$$

So the matrix of multiplication by X is the companion matrix

$$C(f) = \begin{pmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -a_{n-1} \end{pmatrix}.$$

Therefore T is similar to a block diagonal matrix

$$C(f_1) \oplus C(f_2) \oplus \cdots \oplus C(f_t).$$

This is the **rational canonical form**.

Remark 102. Over $F[X]$, the elementary divisors are powers

$$p(X)^e$$

of irreducible polynomials. The summand

$$F[X]/(p(X)^e)$$

gives one block for the linear operator T , because T acts as multiplication by X on this quotient.

If F is algebraically closed, then every irreducible polynomial is linear, so

$$p(X) = X - \lambda.$$

Thus the elementary divisor summands have the form

$$F[X]/((X - \lambda)^e).$$

Set

$$Y = X - \lambda.$$

Then

$$F[X]/((X - \lambda)^e) \cong F[Y]/(Y^e).$$

In this quotient, $Y^e = 0$, so multiplication by Y is nilpotent. Since

$$X = \lambda + Y,$$

multiplication by X has the form

$$T = \lambda I + N,$$

where N is nilpotent.

Using the basis

$$Y^{e-1}, Y^{e-2}, \dots, Y, 1,$$

multiplication by X is represented by

$$J_e(\lambda) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ 0 & 0 & \lambda & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda \end{pmatrix}.$$

This is a Jordan block. Therefore

$$F[X]/((X - \lambda)^e) \longleftrightarrow \text{one Jordan block of size } e \text{ with eigenvalue } \lambda.$$

If F is not algebraically closed, the irreducible polynomials $p(X)$ may have degree greater than 1, so the blocks are companion matrices of $p(X)^e$ rather than Jordan blocks. Thus rational canonical form works over any field, while Jordan canonical form appears when the relevant polynomials split into linear factors.

Algebraic Integers and Character Theory

Definition 103. A complex number α is an algebraic integer if it satisfies a monic polynomial

$$f(X) = X^n + a_{n-1}X^{n-1} + \cdots + a_0 \in \mathbb{Z}[X].$$

Theorem 104. If M is a finitely generated $\mathbb{Z}$ -module and $\alpha M \subseteq M$, then α is an algebraic integer.

Proof. After removing torsion if necessary, one may view the action of α on M as an integer matrix. By Cayley-Hamilton, α satisfies the monic characteristic polynomial of that matrix. $\square$

This is useful in character theory. Let G be finite, let K_j be the conjugacy class of g_j , and set

$$C_j = \sum_{x \in K_j} x \in \mathbb{Z}[G].$$

The class sums multiply as

$$C_i C_j = \sum_k a_{ijk} C_k, \quad a_{ijk} \in \mathbb{Z}.$$

If χ is an irreducible character coming from (π, V) , then $\pi(C_j)$ commutes with $\pi(G)$. By Schur's lemma,

$$\pi(C_j) = \omega_j I.$$

Taking traces gives

$$\chi(1)\omega_j = \text{tr}(\pi(C_j)) = |K_j|\chi(g_j), \quad \text{so} \quad \omega_j = \frac{|K_j|\chi(g_j)}{\chi(1)}.$$

The finite $\mathbb{Z}$ -module spanned by the ω_j is closed under multiplication by each ω_i , so each ω_j is an algebraic integer. Hence

$$\frac{|K_j|\chi(g_j)}{\chi(1)}$$

is an algebraic integer.

Theorem 105. *If χ is an irreducible character of G , then*

$$\chi(1) \mid |G|.$$

Proof. By Schur orthogonality,

$$1 = (\chi, \chi) = \frac{1}{|G|} \sum_j |K_j|\chi(g_j)\overline{\chi(g_j)}.$$

Therefore

$$\frac{|G|}{\chi(1)} = \sum_j \overline{\chi(g_j)} \frac{|K_j|\chi(g_j)}{\chi(1)}.$$

Character values are sums of roots of unity, so they are algebraic integers. Also each

$$\frac{|K_j|\chi(g_j)}{\chi(1)}$$

is an algebraic integer. Hence $|G|/\chi(1)$ is an algebraic integer. It is rational, so it is an integer. $\square$

Theorem 106. *If χ is irreducible and*

$$(|K_j|, \chi(1)) = 1,$$

then either

$$\chi(g_j) = 0$$

or

$$g_j \in Z(\pi),$$

meaning $\pi(g_j)$ acts by a scalar.

Proof. Since $(|K_j|, \chi(1)) = 1$, choose $a, b \in \mathbb{Z}$ such that

$$a\chi(1) + b|K_j| = 1.$$

Then

$$\frac{\chi(g_j)}{\chi(1)} = a\chi(g_j) + b\frac{|K_j|\chi(g_j)}{\chi(1)}.$$

The right side is an algebraic integer, so $\chi(g_j)/\chi(1)$ is an algebraic integer.

But $\chi(g_j)/\chi(1)$ is the average of eigenvalues of $\pi(g_j)$, all of which are roots of unity. If the eigenvalues are not all equal, every conjugate of this average has absolute value < 1 , forcing the algebraic integer to be 0. Thus either $\chi(g_j) = 0$, or all eigenvalues are equal and $\pi(g_j)$ is scalar. $\square$

Final Exam Review

- Know the structure theorem for finitely generated modules over a PID in both forms:

$$R^r \oplus \bigoplus R/(p^e) \quad \text{and} \quad R^r \oplus R/(a_1) \oplus \cdots \oplus R/(a_t), \quad a_1 \mid \cdots \mid a_t.$$

- Be able to prove the three main decomposition steps:

$$M \cong R^r \oplus T(M), \quad T(M) = \bigoplus_p M(p), \quad M(p) \cong \bigoplus_i R/(p^{\lambda_i}).$$

- Be able to pass between elementary divisors and invariant factors using CRT.
- For $R = \mathbb{Z}$, be able to classify finitely generated abelian groups.
- For $R = F[X]$, be able to construct the $F[X]$ -module from a linear map T and recover rational canonical form from invariant factors.
- Know how companion matrices arise from the cyclic modules $F[X]/(f)$.
- Know the algebraic integer criterion:

$$M \text{ finitely generated over } \mathbb{Z}, \quad \alpha M \subseteq M \implies \alpha \text{ is an algebraic integer.}$$

- Know the character theory application:

$$\frac{|K_j| \chi(g_j)}{\chi(1)} \text{ is an algebraic integer}$$

and the proof that

$$\chi(1) \mid |G|.$$

- Review Maschke's theorem, semisimplicity, Schur's lemma, Schur orthogonality, the regular representation, permutation characters, kernels of characters, centers of characters, and how character tables detect normal subgroups.
- Practice constructing character tables, especially for small groups such as dihedral groups and A_4 .
- Review induced characters:

$$(\text{Ind}_H^G \chi)(x) = \frac{1}{|H|} \sum_{g \in G} \chi^0(g^{-1}xg),$$

where χ^0 is the extension of χ by zero outside H .

- Know Frobenius reciprocity:

$$(\text{Ind}_H^G \chi, \psi)_G = (\chi, \text{Res}_H^G \psi)_H.$$

- For Frobenius groups, know the definition, the complement H , the kernel

$$K = \{1\} \cup \{g \in G : g \text{ fixes no point}\},$$

and Frobenius' theorem:

$$K \trianglelefteq G, \quad G = K \rtimes H.$$